Number Theory And Cryptography (MA2209)

ZERO-KNOWLEDGE PROOFS

Project Report

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Abstract

Zero-Knowledge Proofs (ZKPs) are cryptographic protocols that enable one party (the prover) to convince another party (the verifier) that a given statement is true, without conveying any additional information apart from the truth of the statement itself. This project explores the theoretical foundations, mathematical underpinnings, and practical applications of ZKPs. We delve into the core principles of completeness, soundness, and zero-knowledge, and illustrate these through classical examples and modern cryptographic constructs. Real-world use cases, such as secure authentication, blockchain privacy, and electronic voting systems, are discussed to highlight the significance and growing relevance of ZKPs in contemporary digital systems.

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1 Introduction

1.1 Understanding Proofs

In both mathematics and everyday reasoning, the process of **proof** is central to convincing others that a given statement is true. Traditionally, proving a fact involves *revealing information* — you present all relevant details and logical steps that lead to the conclusion.

Example of a Classical Proof

Suppose I claim that the number 26781 is not a prime. To prove this, I could present its factors:

$$26781 = 113 \times 237$$

By verifying this multiplication, you are not only convinced that 26781 is composite — you have also learned its factorization. This is a common characteristic of traditional proofs: they often reveal *more information than necessary* for mere verification.

1.2 Concept of Zero-Knowledge

A typical byproduct of a proof is that you gain some additional knowledge beyond simply being convinced that a statement is true. In the example discussed earlier, not only are you convinced that 26781 is not a prime number, but you also learn its factorization: $26781 = 113 \times 237$.

A **zero-knowledge proof** (ZKP) aims to avoid this. In a zero-knowledge proof, one party (*Alice*) proves to another party (*Bob*) that a certain statement X is true. Crucially, Bob becomes fully convinced that the statement is indeed true, but learns nothing else from the process. In other words, Bob gains *zero knowledge* beyond the truth of the statement.

2 Zero-Knowledge Proofs

Zero-Knowledge Proofs (ZKPs) are interactive protocols that enable a prover to convince a verifier that a given statement is true without conveying any information apart from the fact that the statement is indeed true. The core idea behind ZKPs lies in mathematical logic, number theory, and modern cryptography.
According to **Goldwasser**, **Micali**, and **Rackoff**, who first introduced the "zero-knowledge" concept, there are three main essential properties of zero-knowledge protocols:

1. **Completeness**: If the statement is true and both prover and verifier follow the protocol honestly, the verifier will be convinced of the truth of the statement.

2. **Soundness**: The prover can only convince the verifier of the truth of the statement if that statement itself is actually true, except with some very small probability.

3. **Zero-Knowledge**: The verifier does not know any knowledge (zero knowledge) of the prover's statement or solution that, other than the truth of the statement.

2.1 Mathematical Foundations and Cryptographic Concepts

ZKPs leverage number-theoretic problems that are computationally hard to solve — such as:

- The **Discrete Logarithm Problem (DLP)**
- **Integer Factorization Problem**
- **Graph Isomorphism and 3-Coloring Problems**
- **Elliptic Curve Cryptography (ECC)**

These problems are used because they are:

- **Hard to reverse** without the secret,
- **Easy to compute** in the forward direction.

2.2 Zero-Knowledge Proof for 3-coloring Graphs

Definition 2.1. *A graph G is **3-colorable** if the vertices of the graph can be colored using only three colors such that no two vertices of the same color are connected by an edge.*

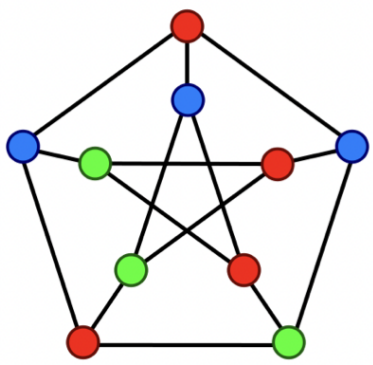

Figure 1: Graph showing 3-colorability.

Let $G = (V, E)$ be a graph. A valid 3-coloring assigns one of three colors to each vertex such that adjacent vertices have different colors. The prover wants to convince the verifier that they know a valid 3-coloring of G without revealing the coloring.

Protocol Outline:

1. The prover randomly permutes the colors and commits to the coloring using a cryptographic commitment scheme (like hash or encryption).
2. The verifier selects a random edge $(u, v) \in E$ and asks the prover to open the commitments for u and v.
3. The prover reveals the colors (after permutation) of u and v, without revealing others.
4. The verifier checks that:
   - u and v are colored differently,
   - the revealed values match the commitments.

Repetition: This protocol is repeated multiple times (say k rounds) to reduce the probability of cheating to 2^{-k}.

Mathematical Insight: The prover never reveals the full coloring — thus, no knowledge is leaked. But by multiple random checks, the verifier becomes confident that a valid 3-coloring exists.

2.3 Discrete Logarithm ZKP (Schnorr Protocol)

Definition 2.2 (Discrete Logarithm Problem (DLP))**.** *Let g be a primitive root for $\mathbb{F}_p$, and let h be a nonzero element of $\mathbb{F}_p$. The* Discrete Logarithm Problem (DLP) *is the problem of finding an exponent x such that*

$$g^x \equiv h \pmod{p}.$$

The number x is called the ***discrete logarithm*** *of h to the base g, and is denoted by $\log_g(h)$.*

Using DLP in Zero-Knowledge Proofs

Zero-Knowledge Proofs enable one party (Prover) to prove to another (Verifier) that they know the solution to a hard problem, such as the discrete log — without revealing the solution itself.

Let G be a cyclic group of prime order q with generator g. Suppose the prover knows x such that $y = g^x \mod p$. They want to prove knowledge of x without revealing it.

Protocol Steps:

1. Prover selects a random $r \in \mathbb{Z}_q$, computes $t = g^r \mod p$, and sends t to the verifier.
2. Verifier sends a random challenge $c \in \mathbb{Z}_q$.
3. Prover computes $s = r + cx \mod q$ and sends s to the verifier.
4. Verifier checks whether $g^s \equiv t \cdot y^c \mod p$.

If the check passes, the Verifier is convinced the Prover knows $\boldsymbol{x}$, but gains zero knowledge of its actual value.

Completeness:

$$g^s = g^{r+cx} = g^r \cdot g^{cx} = t \cdot y^c \mod p$$

Thus, correctness is verified without ever revealing $\boldsymbol{x}$.

2.4 Integer Factorization Problem and ZKPs

Definition 2.3 (Integer Factorization Problem)**.** *The* Integer Factorization Problem *is the problem of finding the prime factors of a given composite integer N. That is, given a positive integer N, determine the prime numbers $p_1, p_2, \ldots, p_k$ and their respective exponents $e_1, e_2, \ldots, e_k$ such that*

$$N = p_1^{e_1} p_2^{e_2} \ldots p_k^{e_k},$$

where p_i are prime and $e_i \geq 1$ for all i.

While factoring small numbers is straightforward, for large integers (hundreds or thousands of bits long), the problem becomes computationally intractable with current classical methods. This computational difficulty is the foundation of various cryptographic systems, such as RSA encryption and digital signatures.

Cryptographic Relevance

The security of the RSA cryptosystem relies on the assumption that factoring a large number N is infeasible. In RSA, the public key includes N, while the factors p and q are kept secret. Knowledge of these factors allows decryption and signing, so protecting them is crucial.

Zero-Knowledge Proofs Using Integer Factorization

Problem: Suppose a prover wants to convince a verifier that they know the factorization of N, but without revealing the actual factors p and q. This is where **Zero-Knowledge Proofs (ZKPs)** come in. The goal is to prove: "I know two primes p and q such that $N = p \cdot q$," without revealing p or q.

Protocol Steps (Blum's ZKP)

- **Commitment (Prover):**
  The prover P randomly selects an element $r \in \mathbb{Z}_N^*$ (i.e., r is invertible modulo N).
  They compute:
  $$x = r^2 \mod N$$
  and send x to the verifier V.
- **Challenge (Verifier):**
  The verifier chooses a random bit $b \in \{0, 1\}$ and sends it to the prover P.

- **Response (Prover):**
  - If $b = 0$: Prover sends r.
  - If $b = 1$: Prover sends $s = r^{-1} \mod N$, i.e., the modular inverse of r modulo N.
- **Verification (Verifier):**
  - If $b = 0$: Verifier checks that
$$x \equiv r^2 \mod N.$$
  - If $b = 1$: Verifier checks that
$$x \cdot s^2 \equiv 1 \mod N.$$

If P knows the factorization of N, they can compute modular inverses in $\mathbb{Z}_N^*$. Otherwise, they cannot respond correctly to both challenges.

2.5 Modular Arithmetic and Its Role in ZKPs

Modular arithmetic, often referred to as "clock arithmetic," forms the backbone of many cryptographic systems including Zero-Knowledge Proofs. In modular arithmetic, numbers wrap around after reaching a certain value called the modulus.

Definition 2.4. *Given a positive integer n, two integers a and b are said to be congruent modulo n if:*
$$a \equiv b \pmod{n} \iff n \mid (a - b)$$
This defines an equivalence relation on $\mathbb{Z}$ and partitions it into equivalence classes modulo n.

Modular Exponentiation and One-Way Functions

A key cryptographic primitive used in ZKPs is the modular exponentiation function:
$$f(x) = g^x \mod p$$
where g is a generator of a cyclic group $\mathbb{Z}_p^*$ and p is a large prime.

This function is easy to compute, but its inverse — solving for x given $g^x \mod p$ — is believed to be hard (Discrete Logarithm Problem), which makes it a one-way function.

Modular arithmetic provides the algebraic foundation for many zero-knowledge proof (ZKP) systems. It is used to define secure, verifiable, and computationally hard mathematical problems that ensure the correctness and confidentiality of the proof.

1. Algebraic Group Structures

Zero-knowledge proofs often operate in cyclic groups such as $\mathbb{Z}_p^*$, where p is a prime. These groups are finite, abelian, and support efficient computation of group operations (multiplication,

inversion), which are necessary for the design of secure protocols.

2. One-Way Functions

Modular exponentiation,

$$y = g^x \mod p,$$

is a one-way function: computing y is efficient, but recovering x (the discrete logarithm) is computationally infeasible. This asymmetry is essential for constructing zero-knowledge challenge–response mechanisms.

3. Hard Problem Construction

Many security assumptions in ZKPs rely on problems defined in modular arithmetic:

- **Discrete Logarithm Problem (DLP)**: Given g, $h = g^x \mod p$, find x.
- **Integer Factorization Problem (IFP)**: Given $N = pq$, find the primes p, q.
- **Quadratic Residuosity Problem**: Given x, determine whether there exists an r such that $x \equiv r^2 \mod N$, without knowing the factorization of N.

4. Commitment Binding

Modular squaring (e.g., $x = r^2 \mod N$) creates a form of computational commitment. While the value x reveals nothing about r, knowledge of r allows the prover to respond to verifier challenges — a cornerstone of zero-knowledge design.

5. Modular Inverses

The ability to compute modular inverses, such as $r^{-1} \mod N$, plays a critical role in protocol verification steps, especially in protocols like Blum's and Schnorr's, where responses depend on multiplicative inverses in $\mathbb{Z}_N^*$.

2.6 Elliptic Curve Cryptography (ECC) and Zero-Knowledge Proofs

Elliptic Curve Cryptography (ECC) is a form of public-key cryptography based on the algebraic structure of elliptic curves over finite fields. ECC provides similar security to traditional methods like RSA but with significantly smaller key sizes, which is particularly beneficial for Zero-Knowledge Proofs where performance is crucial.

Mathematical Foundation of ECC

Definition 2.5. *We define an **elliptic curve over** $\mathbb{F}_p$ to be an equation of the form*

$$E : Y^2 = X^3 + AX + B,$$

where $A, B \in \mathbb{F}_p$ and the discriminant condition $4A^3 + 27B^2 \neq 0$ ensures that the curve is non-singular.

The set of points on E with coordinates in $\mathbb{F}_p$ is denoted by

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p \times \mathbb{F}_p \mid y^2 = x^3 + Ax + B\} \cup \{\mathcal{O}\},$$

where $\mathcal{O}$ is the point at infinity, serving as the identity element for the elliptic curve group.

Group Law on Elliptic Curves

Given two points P and Q on the curve, their sum $R = P + Q$ is defined geometrically:

- Draw the line through P and Q (or the tangent if $P = Q$),
- Find the third point of intersection with the curve,
- Reflect this point across the x-axis to obtain R.

This operation satisfies the group axioms, enabling cryptographic schemes.

ECC-based Zero-Knowledge Proof: Schnorr Protocol on Elliptic Curves

Let $E(\mathbb{F}_p)$ be an elliptic curve, and let G be a base point of large prime order n. The prover knows a secret $x \in \mathbb{Z}_n$ such that:

$$P = xG$$

The goal is to prove knowledge of x without revealing it.

Protocol Steps:

1. Prover picks random $r \in \mathbb{Z}_n$ and computes $R = rG$.
2. Prover sends R to the verifier.
3. Verifier sends a random challenge $c \in \mathbb{Z}_n$.
4. Prover computes $s = r + cx \mod n$ and sends s to the verifier.
5. Verifier checks if:
$$sG \stackrel{?}{=} R + cP$$

Verification:

$$sG = (r + cx)G = rG + c(xG) = R + cP$$

which confirms that the prover indeed knows x, without revealing it.

Why ECC is Suitable for ZKPs

Elliptic Curve Cryptography (ECC) provides a mathematically rich and efficient framework for constructing Zero-Knowledge Proofs (ZKPs). Several core features of ECC make it particularly suitable for this purpose:

1. **Hardness of ECDLP:** The security of ECC is based on the Elliptic Curve Discrete Logarithm Problem (ECDLP), which is computationally hard. Given two points P and $Q = xP$, it is infeasible to compute the scalar x, making ECC an excellent foundation for ZKPs.

2. **Compactness:** ECC achieves high levels of cryptographic security with relatively small key and proof sizes. This is especially beneficial in environments with bandwidth or storage constraints, such as blockchain systems or mobile devices.

3. **Efficient Computation:** Operations on elliptic curves, such as point addition and scalar multiplication, are computationally efficient. This supports fast proof generation and verification.

4. **Group Structure:** The points on an elliptic curve form an abelian group, allowing secure protocols that involve commitments, secret sharing, and homomorphic operations.

5. **Support for Advanced Protocols:** Many modern ZKPs such as Schnorr proofs, Pedersen commitments, Bulletproofs, and zk-SNARKs use ECC as their cryptographic foundation due to its efficiency and security.

In summary, ECC's combination of strong security assumptions, efficient computation, and compact representations makes it a natural and powerful choice for implementing zero-knowledge protocols.

Example: Zero-Knowledge Proof on an Elliptic Curve over $\mathbb{F}_p$

Consider the elliptic curve E defined over the finite field $\mathbb{F}_{751}$:

$$E : y^2 \equiv x^3 + 1 \mod 751$$

Let the base point $G = (0, 1)$ be a known point on the curve of large prime order n. Let the Prover's secret be:

$$x = 113$$

The corresponding public key is:
The public key is:

$$P = xG = 113G = (384, 475)$$

Goal

The Prover wants to convince the Verifier that they know the discrete logarithm x, i.e., that they know the scalar x such that $P = xG$, without revealing x.

Zero-Knowledge Proof Protocol Steps

1. **Commitment (Prover):**
   The Prover selects a random scalar:
   $$r = 98$$
   Then computes the commitment point:
   $$R = rG = 98G = (219, 327)$$
   and sends R to the Verifier.

2. **Challenge (Verifier):**
   The Verifier sends a random challenge:
   $$c = 21$$

3. **Response (Prover):**
   The Prover computes:
   $$s = r + c \cdot x \mod n = 98 + 21 \cdot 113 = 98 + 2373 = 2471 \mod n$$
   The Prover sends s to the Verifier. Assume $n = 757$:
   $$s = 2471 \mod 757 = 200$$
   The Prover sends $s = 200$ to the Verifier.

4. **Verification (Verifier):**
   The Verifier checks whether:
   $$sG \stackrel{?}{=} R + cP$$
   Compute both sides:
   $$sG = 200G = (176, 266)$$
   $$cP = 21 \cdot P = 21 \cdot (384, 475) = (140, 495)$$
   $$R + cP = (219, 327) + (140, 495) = (176, 266) \quad \text{(using elliptic curve point addition)}$$

Since:

$$sG = R + cP = (176, 266)$$

The Verifier accepts the proof.

3 Applications of Zero-Knowledge Proofs

Zero-knowledge proofs, as we have seen, allow a prover to convince a verifier that a statement is true without revealing any other information beyond the validity of the statement. This remarkable property opens up numerous possibilities for enhancing privacy, security, and trust across a wide range of industries. Here are four detailed applications where ZKPs are making a significant impact in real-world systems:

3.1 Privacy-Preserving Cryptocurrency Transactions

One of the most impactful applications of zero-knowledge proofs (ZKPs) is found in the domain of cryptocurrencies, particularly for improving user privacy and confidentiality.

The Problem with Traditional Cryptocurrencies:
Cryptocurrencies like Bitcoin were originally designed to allow anyone to transfer money without needing a trusted central authority. Every Bitcoin transaction is recorded on a public ledger called the blockchain. Although the blockchain does not directly show users' real-world identities, it shows their addresses and the full history of every transaction associated with those addresses. This design means:

- Anyone can see how much Bitcoin is transferred between two addresses.
- Anyone can follow the flow of funds from one address to another over time.
- If an address is ever linked to a real-world identity, all of that user's transaction history can be exposed.

How Zero-Knowledge Proofs Solve This:
Zero-knowledge proofs offer an elegant solution is that they allow a user to prove that a transaction is valid — meaning that:

- The sender actually owns the money.
- The money is not double spent.
- The transaction follows all protocol rules.

Without revealing :

- Who the sender or receiver is.
- How much money is being sent.

- Any other details about the transaction.

In short, validity is proven but the the details remain hidden. This means that the blockchain can still verify transactions and maintain security and integrity, but users retain full privacy over their activities.

A Real Example : Zcash and zk-SNARKs

Zcash is a cryptocurrency that implements privacy using a particular type of zero-knowledge proof called zk-SNARKs. Let's break this down:

zk-SNARK stands for:

- **Zero-Knowledge** : No extra information is leaked.
- **Succinct** : The proof is very small in size.
- **Non-Interactive** : The prover and verifier do not need to go back-and-forth; a single proof is enough.
- **Argument of Knowledge** : It proves that the prover knows something (like a secret key or transaction data) without revealing it.

How Zcash Uses zk-SNARKs When someone makes a transaction in Zcash:

- They create a proof that shows they have enough coins to spend and that the transaction obeys the rules but the proof does not show which coins are being spent or who is receiving them.
- This proof is attached to the transaction and submitted to the blockchain.
- Miners and network nodes can independently verify the validity of this proof without needing to see any private details like sender, receiver, or transaction amount.
- Once verified, the transaction is added to the blockchain as valid.

Thus:

- The blockchain still maintains trust and security (no fake coins or double-spending).
- The users maintain privacy (no public record of their activities).

Visual Summary

Aspect	Bitcoin (Traditional)	Zcash with zk-SNARKs
Transaction details	Fully visible	Completely hidden
Blockchain verification	Publicly checked	Proof checked without details
User privacy	Very low	Very high

3.2 Secure Authentication Systems

Another powerful application of ZKPs is in the area of user authentication — verifying identities securely without revealing sensitive information like passwords.

The Problem with Traditional Authentication:

- Passwords are transmitted (even if encrypted) across networks.
- Servers store passwords (hashed or plain).
- Hacking servers can expose millions of user credentials.
- Phishing attacks easily steal passwords.

How Zero-Knowledge Proofs Solve This:
Zero-knowledge proofs offer a groundbreaking solution where they allow a user to prove that they know the correct password (or secret) without ever sending the password itself. Here's how the idea works:

- Instead of giving the password to the server, the user proves that they know the password through a clever cryptographic challenge-response interaction.
- At no point is the actual password transmitted or even revealed to the server.
- The server only sees the proof, not the password itself.

This dramatically increases the security of authentication.

A Real Example: Passwordless Authentication Protocols
Let's walk through how a zero-knowledge-based login system would work:
1. Setup Phase (When account is created):

- The user chooses a password.
- Instead of storing the password, the server stores a public cryptographic commitment based on the password.
- Think of this like a locked safe that can only be opened if you know the right combination (the password), but the safe itself tells you nothing about the combination.

2. Login Phase (When user tries to authenticate):

- The server sends a random challenge (like a random number or a random puzzle).
- The user uses their password to compute a cryptographic response based on the challenge.
- The user sends back the response.
- The server verifies that the response is correct, based on the public commitment stored earlier.

Visual Summary

Aspect	Traditional Authentication	ZKP-Based Authentication
Password transmission	Sent over network	Never sent
Server stores	Passwords	Cryptographic proofs
Risk if server hacked	Passwords leaked	No password leak
Defense against phishing	Weak	Strong

3.3 Electronic Voting Systems

Electronic voting has the potential to make elections faster, more accessible, and easier to administer. But it comes with a huge challenge: ensuring that votes are counted correctly while maintaining voter privacy. At first, this seems contradictory:

- Verification demands transparency (prove that votes counted right).
- Privacy demands secrecy (keep who voted for whom secret).

The Traditional Problems in Electronic Voting:

- Vote Privacy: risk of tracing votes back to voters.
- Vote Integrity: difficult to verify tally correctness.
- Trust: blind trust required in authorities.

How Zero-Knowledge Proofs Solve This:
ZKPs allow voters to prove their votes are valid and counted without revealing individual votes.

Visual Summary

Aspect	Traditional E-Voting Challenges	ZKP-Based E-Voting Solutions
Voter Privacy	May be compromised	Perfectly preserved
Vote Integrity	Hard to verify	Public, verifiable proofs
Transparency vs. Privacy	Trade-off	Achieve both
Trust Requirement	High	Minimal (math-based trust)

4 Conclusion

ZKPs represent a significant advancement in the field of cryptography, offering a powerful method to verify the validity of information without revealing the underlying data. Through this report, we have explored the theoretical foundations of ZKPs, including interactive proofs, completeness, soundness, and the zero-knowledge property, supported by mathematical tools such as modular arithmetic and elliptic curves. We have also examined real-world applications—ranging from privacy-preserving blockchain protocols to secure authentication —highlighting the growing importance of ZKPs in ensuring both security and privacy in modern digital systems.

As data privacy becomes increasingly critical, ZKPs are poised to play a central role in designing secure, transparent, and efficient systems. Their potential to transform fields like finance, healthcare, and digital governance underscores the relevance of further research and development in this area. In summary, Zero-Knowledge Proofs are not only a theoretical breakthrough but also a practical tool for building trust in an increasingly interconnected world.